

# Sparsh Srivastava

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## Summary

**AI Developer** who has engineered production **AI systems** using **Python**, **LangChain**, and **vector databases**, applying **retrieval-augmented generation** and **large language model** integration across web-connected applications with **PostgreSQL**-backed data design. Benchmarked **1,000+** **AI tools** in a research role and placed **1st** in a **100+** **participant** AI hackathon. Backed by **3** certifications in cloud computing, applied machine learning, and modern web development, with **Git**-driven ownership from prototyping through **cloud deployment**.

## Education

**Vellore Institute of Technology**

Bachelor of Technology, Computer Science & Engineering (AI & ML) | CGPA: **8.05/10.0**

**Sept 2023 – May 2027**

India

**Montfort Inter College**

Class 10 & 12 (Higher Secondary)

**Apr 2021 – Mar 2023**

Lucknow, India

## Technical Skills

**AI/ML:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), LangChain, Prompt Engineering, Model APIs (Groq, Google Generative AI), AI Agent Design, Scikit-learn

**Languages:** Python, Java

**Backend & APIs:** Node.js, REST API Development & Design, Authentication

**Frontend:** React.js, Next.js, HTML5, CSS3

**Cloud & Databases:** PostgreSQL, Supabase (DB & Auth), Database Design, Render Deployment

**Tools & Practices:** Git, GitHub, Dataset Development

## Work Experience

**Future Smart AI**

**Sept 2025 – Feb 2026**

Content Research Intern | [Link](#)

Remote

- Evaluated and benchmarked more than **1,000 AI models and tools** across **4 categories** – video, image, captioning, and content generation – against marketing claims, producing a trusted, evidence-based capability assessment.
- Authored comparative technical reports scoring each model against **3 criteria** – strengths, limitations, and best-fit use case – helping teams select the right AI solution faster and with greater confidence.

## Projects

**Interview IQ** | [Live Demo](#) | [GitHub](#)

**Jul 2025 – Aug 2025**

- Candidates lacked grounded prep – architected a document-grounded **RAG** pipeline with **LangChain** and a **vector database** for context-aware retrieval – delivered verified, source-backed answers.
- Prep felt generic – engineered a **PostgreSQL**-backed data layer logging history across **4 activity types** – tailored each candidate's study path with real signals.
- Prep resources were scattered – delivered **3 formats** of **LLM**-grounded study material, **VAPI**-powered voice-AI interviews, and a resume gap analyzer – unified prep into one workflow.

**AI-EchOo E-Commerce Platform** | [Live Demo](#) | [GitHub](#)

**Feb 2026**

- Shoppers faced a static catalog – built a **generative-AI** shopping assistant across **6 purchase flows** – turned browsing into guided discovery and comparison.
- Assistant answers needed grounding – designed a relational **4-table** schema and integrated **Groq**-hosted **LLMs** with **RAG** – grounded responses in live inventory.
- Assistant sometimes invented answers – eliminated fabricated recommendations by anchoring suggestions to real-time stock data – fixed sync issues, raising reliability at checkout.

**Desktop AI Voice Assistant** | [Live Demo](#) | [GitHub](#)

**May 2025**

- Manual input slowed tasks – developed a Python-based **AI agent** wiring **Groq** and **Google Generative AI** into **REST API** calls – enabled hands-free voice and text commands.
- Assistant felt robotic – implemented two-way speech-to-text/text-to-speech via a **PyQt5** interface with **asynchronous processing** – automated browser workflows across **2 tasks**.
- Commands took multiple steps – optimized the inference pipeline to execute **4 categories** of commands – collapsed steps into one spoken instruction.

## Achievements

**Won 1st Place** among **100+** **participants** at the AI Demos Monthly Hackathon for an innovative AI solution. ([Link](#))

**Published a Carbon Credits Dataset** on Mendeley Data for a machine learning-based carbon credit prediction model.

([Dataset](#) | [Predictor](#))

## Certifications

- Microsoft Azure Fundamentals (AZ-900)** – Cloud computing, cloud services, security fundamentals ([Certificate](#))
- Applied Machine Learning in Python** – University of Michigan, Coursera ([Certificate](#))
- MERN Stack** – MongoDB, Express.js, React.js, Node.js ([Certificate](#))